

Security Challenges in AIoT Systems



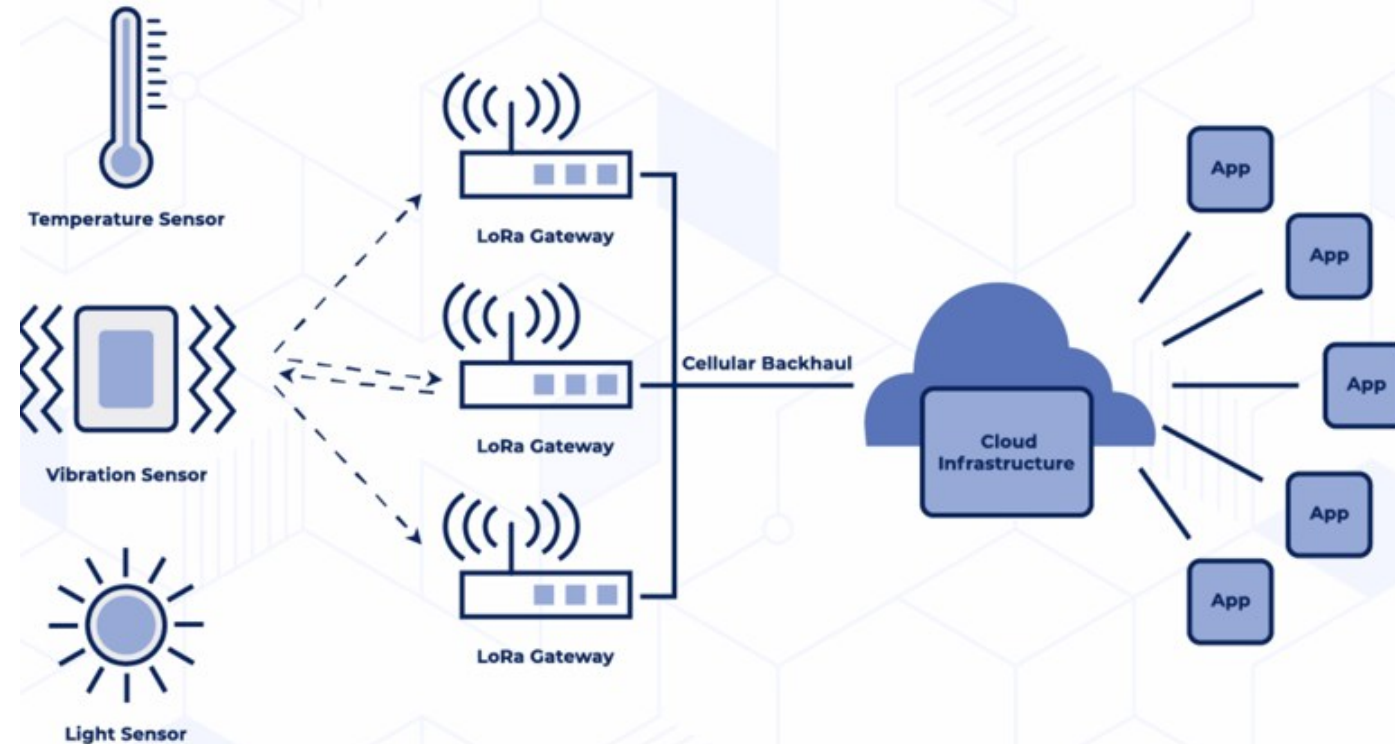
Why AIoT Requires a New Security Mindset ?

- Connected smart devices
- Distributed intelligence (Edge AI)
- Cloud platforms & analytics
- Autonomous decision-making

Unlike traditional IT:

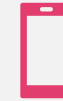
- Devices are always on
- Often unmonitored
- Deployed in untrusted locations
- Updated infrequently

IoT Threat Landscape



What Is an IoT Attack Surface?

- The **IoT attack surface** is the *sum of all points where an attacker can enter, manipulate, or disrupt an IoT system*—from physical devices to cloud APIs.



1. Device Layer



2. N/W Layer



3. Gateway / Edge Layer



4. Cloud & Backend Layer



5. Application & User Layer

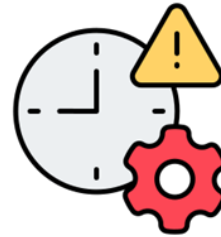
Core Security Challenges



**Weak
Authentication &
Default
Passwords.**



**Inadequate
Update
Mechanisms.**



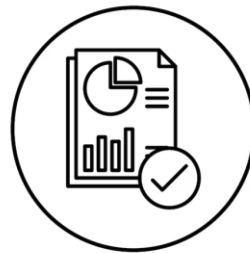
**Resource
Constraints.**



**Data Privacy &
Encryption Gaps.**



**Physical
Exposure &
Tampering.**



**Lack of
Standardization.**



**Large,
Distributed
Attack Surface.**

Core AIoT Vulnerabilities

The integration of AI into IoT introduces specific, high-impact risks that traditional networks do not face:



Adversarial AI Attacks: Attackers can manipulate input data (e.g., sensor data or camera feeds) to "trick" AI models into making incorrect autonomous decisions, such as misidentifying a threat in a surveillance system.



Resource Constraints vs. Security: Many AIoT edge devices have limited processing power and battery life, making it difficult to run heavy encryption or advanced Intrusion Detection Systems (IDS).



Model Inversion & Data

Poisoning: Malicious actors may attempt to steal the underlying AI model (intellectual property) or "poison" the training data to create backdoors for future exploitation.



Supply Chain Weakness: Vulnerabilities often cascade through the ecosystem, starting from compromised chips or default credentials in third-party hardware.

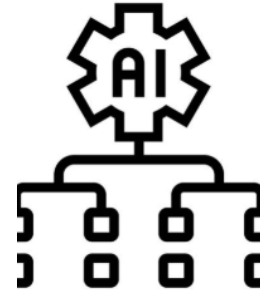
Emerging Threat Trends



Automated Botnets: Attackers are using AI to manage massive IoT botnets, allowing them to conduct targeted ransomware assaults that can infiltrate critical networks in under 24 hours.



Cloud & Hybrid Risks: As data moves between edge devices and the cloud, single-factor authentication and unencrypted transmissions remain primary targets for data breaches.



Agentic AI Threats: The rise of autonomous "agentic AI" allows threat actors to delegate complex tasks to AI agents, though this is currently considered a secondary priority compared to immediate phishing and malware threats.



OT/ICS Targeted Attacks: Operational Technology (OT) and Industrial Control Systems (ICS) are increasingly vulnerable due to wireless connectivity "blind spots" that attackers use for deauthentication and data modification.

Strategic Defenses & Frameworks



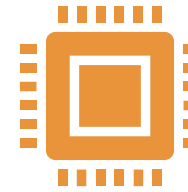
Zero-Trust Architectures

Implementing Zero-Trust principles ensures that no device, even internal, is trusted by default.



Edge Computing Security

Processing data locally at the edge reduces the risks associated with data transit to the cloud, though it requires specialized, lightweight security protocols.



Human Firewall

Despite automation, human intuition remains the final defense against AI-driven phishing and deepfakes.



Regulatory Compliance

Following guidelines from agencies like ENISA and NIST is becoming essential for ensuring security throughout the product lifecycle.